Unsupervised Hyperspectral Band Selection via Hybrid Graph Convolutional Network

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***Abstract*— Hyperspectral image (HSI) provided with a substantial number of correlated bands causes calculation consumption and an undesirable "dimension disaster" problem for the classification. Band selection (BS) is an effective measure to reduce the information redundancy with the physics spectrum preserved for HSI. Although the existing BS methods have achieved noticeable progress, the correlation between neighbor bands still needs to be mined deeply for an effective selection criterion. This article proposes a BS approach to collecting the discriminative band subset for HSI classification (HSIC), which adopts the self-supervised learning paradigm to implement the BS by the auxiliary spectrum rebuilding (SR) task. In specific, we utilized a convolutional neural network (CNN) and a graph convolutional network (GCN) for the spectral–spatial feature extraction. Next, GCN and CNN are developed for the refinement of the band correlation sequentially. Afterward, the selected bands in terms of the acquired correlation are fed into the presented self-supervised SR network for spectral reconstruction. Simultaneously, the proposed architecture completed the selection with the optimization of the band reconstruction by a defined loss function. In this way, we supply substitution for selection criterion and path searching through the end-to-end framework. The extensive experimental results and analysis demonstrated that the proposed hybrid architecture provided a competitive band subset for the classification, and the accuracies with different types of classifiers are more effective than the compared BS methods.**

***Index Terms*— Band selection (BS), convolutional neural network (CNN), graph convolutional network (GCN), hyperspectral image classification (HSIC), spectrum rebuilding (SR).**

I. Introduction

HYPERSPECTRAL remote sensing technology [1]–[7] can collect the spectral response information of ground objects from visible light to near-infrared in tens to hundreds of narrow bands and distinguish ground objects with subtle spectral differences through classification. With the rapid development of hyperspectral remote sensing technology, it has been widely applied to various applications, such as military [8], [9], medical treatment [10]–[13], water quality monitoring [14], and agricultural monitoring [15], [16]. However, the continuity of hundreds of spectral bands causes the adjacent bands to be highly correlated, which brings a certain degree of redundancy and increases the computational burden. Therefore, decreasing the redundancy of hyperspectral data is indispensable for hyperspectral image classification (HSIC) [17]–[21] and target detection. Nowadays, band selection (BS) is one of the effective methods to alleviate this problem, which refers to retaining the most representative or informative bands from original spectral bands.

Generally speaking, the main mechanisms of existing BS methods consist of supervised, semisupervised, and unsupervised approaches. The process of supervised learning is about adjusting the parameters of the model to achieve the desired performance by using a group of samples of known labels. Unsupervised learning is a machine learning (ML) mechanism [17] without a training set, which can find potential patterns in datasets without labels; some representative unsupervised algorithms include clustering, discrete point detection, and dimensionality reduction. Due to the high cost of data-labeling, semisupervised methods extract information with labeled and unlabeled samples simultaneously.

Research on the supervised-based BS method is long-standing and tends to be mature in the HSI recognization area. Patra *et al.* [18] defined a new standard to select information bands with higher relevance and meaning values after calculating the correlation and saliency of each spectral band by the rough set theory. In recent years, with the continuous development of deep learning (DL) technology, the BS method based on DL is gradually matured and widely used. Zhan *et al.* [19] proposed a new CNN approach based on distance density (DDCNN) for BS, in which the band subset is selected by measuring the distance density between the spectral bands; different classifiers are used to measure the classification effect of the selected band subset finally. However, the existing DL methods usually perform BS and HSIC separately or evaluate the selected spectral bands by training the deep network iteratively, which may lead to the loss of

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identified bands and an increase in computing cost. In [20], a novel CNN based on bandwise independent convolution and hard thresholding (BHCNN) is presented, which combines BS, feature extraction (FE), and classification in a whole framework, where the BS layer performs for each spectral band of input HSI independently. Then, hard thresholding is utilized to restrain the weights of the convolution kernels with the unselected bands to zero.

The previous semisupervised method was a preliminary exploration of the value of unlabeled samples in the traditional supervised method [21]. The semisupervised approach for BS has attracted a great deal of attention over the last few decades due to the lower cost of data sampling and label annotation. Guo *et al.* [22] proposed a semisupervised method based on the hypergraph model to select productive bands. The method starts with constructing a hypergraph that contains all the feature points in HSI and learns the projection matrix in a semisupervised manner. The minimum angle regression algorithm is used to solve the minimum problem and select the optimal frequency band. Feng *et al.* [5] designed a low-order approximation criterion based on high resolution, high information, and low redundancy. From the perspective of the information theory, it is proven that many traditional semisupervised feature selection criteria are low-order approximations of the maximum discrimination and information (MDI) criteria. The experimental results on HSI show the effectiveness of the semisupervised BS method. In [23], a semisupervised wrapper frequency BS method based on dynamic classifier selection is proposed. Specifically, the support vector machine (SVM) classification map is provided with a high-quality reference, utilizes the k-nearest neighbor method to define the local area, and, finally, chooses the bands with the best classification performance. Feng *et al.* [24] present a semisupervised BS method based on reinforcement learning (RL), which regards band search as a sequential decision-making process and describes each state in the search space by constructing EvaluateNet; moreover, the network creates a random band sampling method to train EvaluateNet that can be evaluated efficiently without any fine-tuning. Finally, the superior evaluator algorithm is designed to explore the state space and select a subset of bands based on the desired cumulative reward. Unlike traditional ML models and CNN approaches, the emergence of GCN breaks the situation that these methods can only deal with regular data; GCN operates on graph-structured data. To better handle the irregular class boundaries, the recently proposed graph convolutional network (GCN) has been adopted for HSIC [25]–[29] and HSI BS. Feng *et al.* [30] propose a novel dual-GCN based on band attention and a sparse constraint for BS by using minibatches of labeled and unlabeled samples to construct a graph. The network consists of two branches that are the attention branch and the trunk branch. In the attention branch, band-based dual graphs are constructed to encode the contextual correlations of adjacent bands and structural correlations of long bands into non-Euclidean spaces. In the trunk branch, sample-based dyadic graphs are constructed to represent the topological information of the samples in the spectral and spatial domains.

State-of-the-art methods for BS of hyperspectral images (HSIs) in an unsupervised manner are basically of two types: ranking-based methods and clustering-based methods. Some ranking-based methods for HSI, which are presented in the literature, are the information divergence (ID)-based methods [31]. Martinez-Uso *et al.* [32] propose a BS method based on a hierarchical clustering structure to group bands. Qian and Yao [33] perform clustering over bands to group them according to their redundancy and select one representative band from each cluster of bands. In [34], an unsupervised BS method that combines clustering and sorting is proposed and achieves promising results compared to other BS approaches.

Although many methods select a compact and low redundant band subset, which is inadequate as it may degrade the classification performance, Sui *et al.* [35] proposed a robust unsupervised BS method that attaches importance to select representative bands; this method revealed bandwise representation based on the comprehensive interband neighborhood structure. In order to ensure the retention of bands involving abundant intrinsic structures conducive to classification, in [36], a manifold-preserving band-importance metric is put forward to measure the bandwise essentiality; it is conducive to classification with the retention of bands involving abundant intrinsic structure. Genetic algorithms provide high-quality solutions to various problems involving search optimization and learning. Zhao *et al.* [37] propose an unsupervised BS method based on improved Genetic algorithms that generate a fitness function based on the Fisher score combined with superpixels constructed to evaluate the discriminability of the band subset considering spectral and spatial information; the method implements an improved genetic operation to restrain the search space and reduce the redundancy.

Inspired by the idea of column subset selection in numerical linear algebra, Wang *et al.* [38] proposed an unsupervised BS algorithm for HSI based on the column subset selection problem in numerical linear algebra. The algorithm utilizes the Manhattan distance to obtain a higher quality of the selected subset with maximum volume. In [39], two methods were proposed, which aim to select different information features in the prototype space (PS) through geometric interpretation of distinct bands. The first method selected the initial feature based on the orthogonal distance of the PS diagonal and utilizes the HySime algorithm in PS to determine the optimal feature size. The second method utilized the angle tangent between two standard defined band vectors to select the initial feature and describe the band correlation. Yu *et al.* [40] built a new class signature-constrained background suppressed approach BS method, which can minimize the influence of unknown BKG and improve classification performance. Many existing BS methods estimate the saliency of each band separately but cannot fully consider the nonlinearity and global interaction between spectral bands. Recently, an end-to-end BS framework is proposed, which implements two different BS-Nets [41], including the fully connected networks (BS-Net-FC) and the convolutional neural networks (CNN) (BS-Net-Conv), based on different application scenarios. The BS-Nets focus on the selection criterion design by band reconstruction task, and the top-k bands are selected eventually after the network

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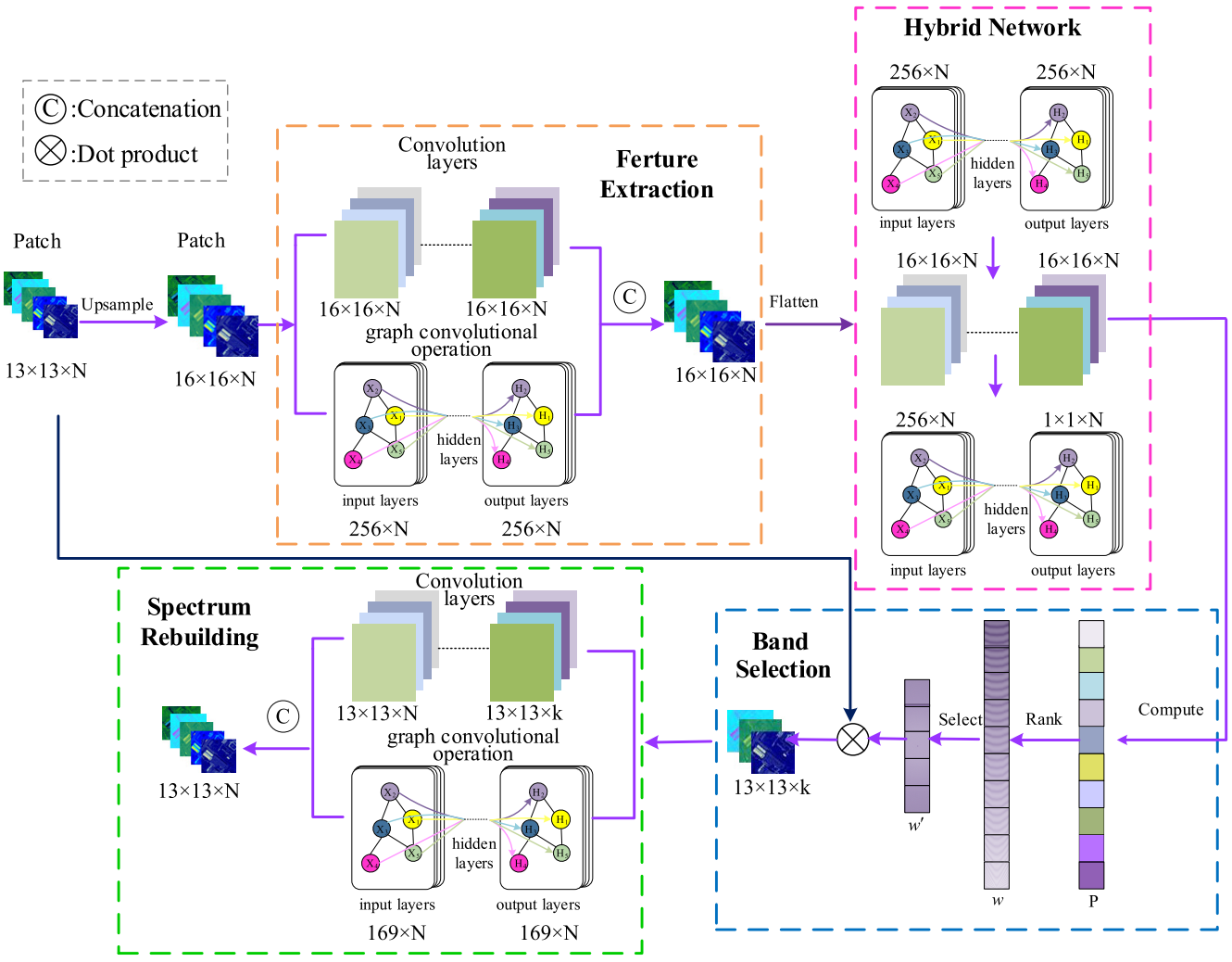


Fig. 1. Structure diagram of the proposed BS framework. Specifically, the FE module is responsible for learning the features of each band. Next, a hybrid network is developed for the refinement of the band correlation sequentially, and top-ranked spectrum bands are chosen by the BS module. Afterward, the selected bands are fed into the SR module for the optimization of BS.

training, which may lose some identifiable bands due to the separation of selection and evaluation.

This article aims at selecting distinct and informative features in irregular spaces to improve the classification accuracy of HSI. Our proposed approach adopts the self-supervised learning paradigm to implement the BS by auxiliary spectrum rebuilding (SR) task and supplies substitution for selection criterion and path searching through an end-to-end framework. Instead of the difficulty of construct selection measurement, the GCN hybrid model is presented to mine the valuable band correlation automatically. SR as a self-supervised learning task substitutes for the complex searching of subset paths. On the one side, the GCN block expands the band correlation by utilizing the non-Euclidean domain information. On the other side, the self-supervised learning scheme implements the BS in a way that the accomplishment of the auxiliary SR task implies that the input bands of the encoder network are exactly the signals to be selected.

In specific, we first utilized a CNN and GCN (GCN-CNN-BS) for the spectral–spatial FE, where GCN and CNN are developed for the refinement of the band correlation sequentially. Next, the selected bands in terms of the calculation with the acquired correlation are fed into the presented reconstruction network for SR. Subsequently, the proposed architecture completed the selection with the optimization of the band reconstruction by a defined loss function.

The primary contributions of this article can be summarized as follows.

1) We presented a self-supervised learning paradigm to implement the BS by auxiliary SR task, which supplies substitution for selection criterion and path searching through an end-to-end framework. The proposed framework is the first attempt to develop a self-supervised paradigm for BS of HSI imagery to the best of our knowledge. The advantage of the self-supervised pattern performs substitution for traditional subset searching and evaluation.

2) We developed a hybrid extraction module that integrates both GCN and CNN in a whole framework to enable the network for information refinement. In which, the GCN block expands the band correlation by utilizing the

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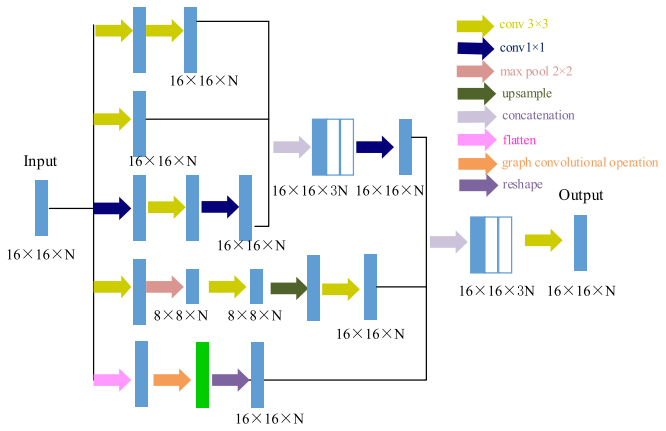


Fig. 2. Exhibition of the basic module of the FE module and the SR module that are designed with the same structure except for the different input dimensions. Specifically, the FE is composed of two basic modules, and the SR consists of four basic modules. The inputs of the two networks are the original band number and the BS number, respectively.

non-Euclidean domain information, which is superior to the previous approaches.

3) We conducted experiments on four different types of classifiers with state-of-the-art BS methods to demonstrate the effectiveness of the proposed approach comprehensively.

The rest of this article is organized as follows. Section II introduces our proposed architecture in detail. In Section III, the experimental results and analysis based on benchmark HSI datasets are presented. Conclusions are drawn in Section IV.

II. Proposed Method

As illustrated in Fig. 1, the proposed method contains four parts: the FE module, the hybrid network for feature fusion, the BS block, and spectrum reconstruction. The individual steps of the FE module and the SR module are presented in Fig. 2. They are designed with the same structure with the obvious difference in the dimensions of the input of the two modules. Specifically, the original HSI is fed into the FE module, and the purpose of this module is to extract the spatial and spectral information of the HSI. The selected bands are chosen through the hybrid network and the reconstruction network, which aims to reconstruct the input selected subset into the initial HSI. The components of the framework are described in detail in the following.

A. Graph Representation of Hyperspectral Bands

Assume that the graph $G = (\mathrm{V}, \mathbf{A}, E)$ consists of a vertex set V and edge set E, where $\mathrm{V} = \{\mathrm{v}_1, \mathrm{v}_2, \ldots \mathrm{v}_\mathrm{n}\}$ is a set of n vertices, E represents the edge relationship on a graph, if v_i and v_j are connected, which means that (v_i, v_j) form an edge $(v_i, v_j) \in$ E, $\mathbf{A} \in R^{\mathrm{N}\times\mathrm{N}}$ is the adjacency matrix, and v_{ij} is an edge between vertex i and vertex j. Owing to the nonuniformity of the bands of HSI, invariance of arrangement, and extra edge properties of the graph, the CNN on the specification grid cannot be directly applied to the graph. GCN works by aggregating features between nodes based on the adjacency matrix $\mathbf{A}$, which cannot be directly applied to HSI due to the unclear relationship between nodes and edges.

GCN is a novel neural network that works by aggregating immediate neighbor features and combining them with the features of the nodes. In this article, we make an effort to convert the HSI into a topological structure. To be specific, the main idea of the method aims to generate graph structure by assuming each band of the HSI as a node to capture the dependencies between bands. The establishment of the adjacency matrix plays a crucial role in the construction of graph structure. Since the calculation of the similarity between bands to generate the adjacency matrix $\mathbf{A}$ causes more time cost, the neural network needs to pay more attention to learning how to construct the adjacency matrix in the process of backpropagation, and the ability of the neural network to learn band features will be weakened in this way. Moreover, the variability of similarity constructed from the band features is limited to a small range, which increases the difficulty of setting the similarity threshold. To further improve the difference in each generated graph structure and fully capture dependencies between nodes and edges, a Bernoulli distribution obeying α is utilized for random sampling to construct the adjacency matrix of the graph, which is responsible for the generation of a random topological structure. The generation process of the adjacency matrix $\mathbf{A}$ can be defined as

$$\mathbf{A}_{ij} = \begin{cases} 1, \left(r_{(v_i,v_j)} = 1\right) \\ 0, \left(r_{(v_i,v_j)} = 0\right) \end{cases}, r_{(v_i,v_j)} \sim \text{Bernoulli}(\alpha), v_i, v_j \in \mathrm{V} \tag{1}$$

where $r_{(v_i,v_j)} \sim$ Bernoulli(α) indicates that $r_{(v_i,v_j)}$ obeys the Bernoulli distribution with the parameter of $\alpha, r_{(v_i,v_j)} = 1$ means that v_i and v_j are connected, and $r_{(v_i,v_j)} = 0$ means that v_i and v_j are not connected. In specific, we obtained $r_{(v_i,v_j)}$ in a style of random generation way with the ratio of α, which is fixed to 10% in the implementation. With the increment of α, the connection between nodes becomes denser.

B. Feature Extraction Module

To fully extract feature information of HSI, we designed the FE module based on the inception network, as shown in Fig. 2. On the whole, the FE module employs two layers of basic module for FE, the initial input of patch size is (13, 13, N), where N is the number of HSI bands, and then, the size is fixed by upsampling (bilinear interpolation) of (16, 16, N) after using the basic module for the first time, subsequent FE is carried out after further utilize basic module, and the final output size is (16, 16, N).

Specifically, one part is the CNN-based model with different convolution kernel sizes of 1×1 and 3×3 to capture the potential local features of HSI. In the other part, in order to fully consider the nonlinear and global interactions between spectral bands, the module incorporates GCN that serves to acquire nonlocal information of HSI. The adjacent matrix of GCN is constructed based on (1) and the sampling probability α, where the node represents the bands, and the edge of the

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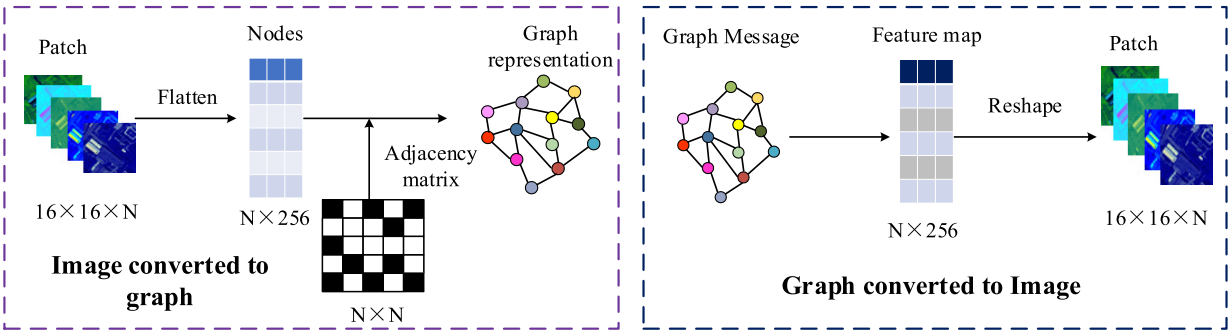


Fig. 3. Illustration of the feature transformation between the GCN module and the CNN module.

node represents the distance between nodes. GCN can work on the graph-structured non-Euclidean data that collectively aggregate the node features in a graph in a global mechanism.

For the graph G, the degree matrix of the adjacency matrix is $\mathbf{D}$, which is a diagonal matrix. The Laplace matrix of the graph G can be defined as $\mathbf{L} = \mathbf{D} - \mathbf{A}$, and the symmetric normalized Laplace matrix is defined as

$$\bar{\mathbf{L}} := \mathbf{I}_N - \mathbf{D}^{-\frac{1}{2}}\mathbf{A}\mathbf{D}^{-\frac{1}{2}} \tag{2}$$

where $\mathbf{I}$ is the identity matrix. The spectral convolution of signal $x \in R^N$ $g_\theta = \mathrm{diag}(\theta)$ and filter with a parameter of $\theta \in R^N$ in the Fourier domain are shown as follows:

$$g_{\theta'} * x \approx \theta\left(\mathbf{I}_N + \mathbf{D}^{-\frac{1}{2}}\mathbf{A}\mathbf{D}^{-\frac{1}{2}}\right)x \tag{3}$$

where $\mathbf{I}_N + \mathbf{D}^{-\frac{1}{2}}\mathbf{A}\mathbf{D}^{-\frac{1}{2}}$ is an eigenvalue with a range [0,2]

The GCN learns the spatial context information and fully extracts the interdependences between the spectral bands. The inputs of the GCN layer contain the adjacent matrix $\mathbf{A}$ and node feature. The formulation of the graph convolutional layer is expressed as

$$\mathrm{H}_{\mathrm{GCN}}^{(\ell)} = \delta\left(\tilde{\mathbf{D}}^{-\frac{1}{2}}\tilde{\mathrm{A}}\tilde{\mathbf{D}}^{-\frac{1}{2}}\mathrm{H}_{\mathrm{GCN}}^{(\ell-1)}\mathbf{W}_{\mathrm{GCN}}^{(\ell)}\right) \tag{4}$$

where $\mathbf{A}^{\sim} = \mathbf{A} + \mathbf{I}_{\mathrm{N}}$, $\tilde{\mathbf{D}}_{ii} = \sum_{j=1}^{N}\tilde{\mathrm{A}}_{ij}$, $\tilde{\mathbf{D}}^{-(1/2)}\tilde{\mathrm{A}}\tilde{\mathbf{D}}^{-\frac{1}{2}}$ is the renormalized form of $\mathbf{D}^{-(1/2)}\mathbf{A}\mathbf{D}^{-(1/2)}$, $\mathrm{H}_{\mathrm{GCN}}^{(\ell-1)}$ represents the input of the ℓth layer, $\mathrm{H}_{\mathrm{GCN}}^{(\ell)}$ denotes the output of the ℓth layer, $\mathbf{W}_{\mathrm{GCN}}^{(\ell)}$ is the trainable weight matrix involved in the ℓth layer, and δ represents an activation function that is implemented by rectified linear units (ReLUs) in our method.

Notably, the conversion dimension transmission is indispensable for the GCN and CNN modules in the hybrid network. Specifically, since the feature of each graph node is represented by a 1-D vector, we use the operation of flatten to convert the feature map of size (h, w, N) to the vector of dimension $(1, h \times w)$ fed into the GCN part. Subsequently, the operation of reshaping is adopted to transform the vector of dimension $(1, h \times w)$ to the feature map of size (h, w, N), which denotes the data of one node of the N channels for the following CNN module.

In particular, we utilize concatenation that is defined as (5) to fuse features obtained by the CNN-based model and GCN-based model, such as DenseNet

$$\mathrm{H}_{\mathrm{CNN}}^{(\ell)} = \mathrm{H}_\ell\left(\left[\mathrm{H}_{\mathrm{CNN}}^{(\ell-1)}, R\left(\mathrm{H}_{\mathrm{GCN}}^{(\ell-1)}\right)\right]\right) \tag{5}$$

where $[\mathrm{H}_{\mathrm{CNN}}^{(\ell-1)}, R(\mathrm{H}_{\mathrm{GCN}}^{(\ell-1)})]$ refers to the concatenation of the feature maps produced in the $(\ell - 1)$th layer, and H_ℓ represents a composite function of three consecutive operations: batch normalization (BN), followed by an ReLU, and a 3×3 convolution (Conv).

Consequently, the concatenate layer fuses the semantic information of feature maps at different scales and increases the channel information, which can feed richer information into subsequent models for improving the overall performance of the model. Therefore, the concatenate layer can provide richer information for the subsequent model. By combining CNN and GCN, the receiving domain increases with the detailed texture features.

C. Hybrid Network for Band Correlation Mining

In the hybrid network, we stack the three layers of CNN and GCN to aggregate and update the information of nodes. GCN aims to analyze data by learning information transfer patterns between graph structures and discover potentially complex structures in the data. First, the hybrid network generates a graph to convert the output of the FE module into graph data. Subsequently, we adopt a GCN layer with feature descriptions $\mathbf{X}$ and the adjacency matrix $\mathbf{A}$ as input to extend the BS problem into the non-Euclidean domain, where input and output sizes are $(16, 16, N)$. The output of a one-layer GCN can be mathematically expressed as (4).

Different from the traditional two-layer GCN network, the hybrid model adds several layers of CNN between two-layer GCNs. The transformation process between image and graph is shown in Fig. 3. The output of the GCN is fed to CNN, which adopts maximum pooling to facilitate downsampling, the output size is $(8, 8, N)$, and the process of downsampling aims to obtain more accurate global information of the bands. The current image is transformed to the size of $(16, 16, N)$ by using upsampling, which aims to make the bilinearly interpolated data reflect more information about the bands. The input of the CNN layer is obtained by the output of one-layer GCN, which is formulated as

$$\mathrm{H}_{\mathrm{HYBRID}}^{(\ell)} = \delta\left(\mathrm{H}^{(\ell-1)}(x; \mathbf{W}_{\mathrm{CNN}}) + b\right) \tag{6}$$

where $\mathbf{W}_{\mathrm{CNN}}$ and b indicate weights and bias, respectively. $\delta(.)$ represents an activation function that is implemented by the ReLU function in our model, $\mathrm{H}^{(\ell-1)}$ indicates the information

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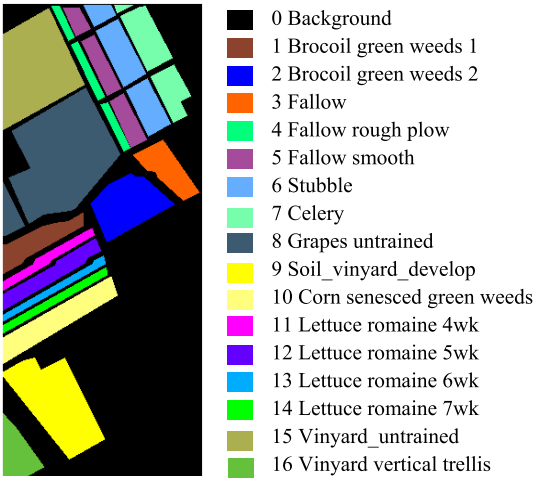


Fig. 4. Colorful ground-truth image of salinas.

obtained by the front GCN module, and $\mathrm{H}_{\mathrm{HYBRID}}^{(\ell)}$ denotes the feature acquired by the GCN and CNN layers.

Furthermore, we aggregate the high-level features from the neighbors by GCN, the input size is $(16, 16, N)$, the output size is $(1, 1, N)$, and the propagation rule of the simplified GCN model is defined as

$$\mathrm{H}^{(\ell+1)} = \delta\left(\tilde{\mathbf{D}}^{-\frac{1}{2}}\tilde{\mathrm{A}}\tilde{\mathbf{D}}^{-\frac{1}{2}}\mathrm{H}_{\mathrm{HYBRID}}^{(\ell)}\mathbf{W}_{\mathrm{GCN}}^{(\ell)}\right) \tag{7}$$

where $\mathrm{H}_{\mathrm{HYBRID}}^{(\ell)}$ is not only the output of the ℓth layer but also the input of the $(\ell+1)$th layer, and $\mathbf{W}_{\mathrm{GCN}}^{(\ell)}$ is the weight matrix of trainable parameters at the ℓth layer of GCN.

In this way, GCN can update the state of a node by gradually aggregating information in the neighborhood to represent the complex dependencies between nodes, which allows us to efficiently capture dependencies between bands.

D. Band Selection Based on SR

The combination of CNN and GCN enhances the ability to distinguish band features, which is beneficial to selecting discriminative bands. In the proposed framework, the hybrid network is followed by the Softmax activation function, which calculates the probability values Pto generate the band correlation coefficient w. We select the k largest values after sorting them as the spectral reconstruction coefficient w'. The following formula shows the reweighted band as

$$X = x \otimes w' \tag{8}$$

where X is the result of reweighting the input x and $\otimes$ indicates the bandwise production between x and w'.

Reconstructing the initial HSI by utilizing a small amount of HSI is a crucial step in our architecture, and we eventually select the subset that can maintain the stability of the network. To be specific, the selected bands in terms of the calculation with the acquired correlation are fed into the presented SR module with a similar structure as the FE module for SR, where the input size is $13 \times 13 \times k$ and the output size is $13 \times 13 \times N$. Subsequently, the proposed architecture completed the selection with the optimization of the band reconstruction

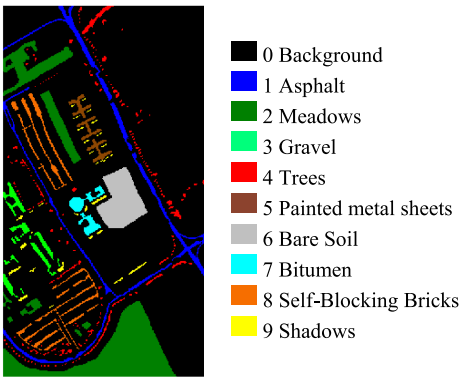


Fig. 5. Colorful ground-truth image of pavia university.

by a defined loss function, which is designed to detect the difference between the reconstructed image and the original image

$$\mathrm{Loss} = \frac{1}{s}\sum_{i=0}^{s}\|x_i - \hat{x}_i\|_2 \tag{9}$$

where S is the total number of training samples, and x_i and $\hat{x}_i$ are the ith original and reconstruction pixels, respectively.

The implementation details of GCN-CNN-BS are shown in Algorithm 1.

Algorithm 1 GCN-CNN-BS Method

Input: HSI cube: $X \in \mathrm{R}^{N\times M\times b}$
Output: Band index.
Initialize: The number of selected bands k,α and the learning rate.
1 while *Model does not converge or satisfy the maximum iteration*
do
2 Construct adjacency matrix **A**by Eq.(1);
3 Train GCN and CNN with the input$X$;
4 Calculate bands weights w;
5 Sort the weights w with k maxima reserved;
6 Re-weight spectral bands by Eq. (8);
7 Reconstruct spectral bands;
8 Calculate the loss of reconstructed bands by Eq. (10);
9 Update GCN and CNN parameters using Adam optimizer;
10 end
11 Select the top k bands.

III. Experiments and Result Analysis

A. Hyperspectral Datasets' Description

The performance of the proposed method is evaluated on three datasets, i.e., the Salinas, the Pavia University, and the Houston, which will be introduced in the following.

1) Salinas Dataset: The Salinas dataset was collected by the 224-band AVIRIS sensor over Salinas Valley, California; it was characterized by high spatial resolution (3.7-m pixels). The dataset comprises 512×217 pixels and 16 classes.

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TABLE I
ARCHITECTURE OF OUR PROPOSED GCN-CNN-BS (THE BASIC MODULE IS SHOWN IN FIG. 2)

Stage	Input Size	Operation	Output Size
Feature extraction	$16\times16\times N$	basic module ×2	$16\times16\times N$
Hybrid network	$16\times16\times N$	GCN	$16\times16\times N$
	$16\times16\times N$	CNN ×2	$16\times16\times N$
	$16\times16\times N$	2×2 max pool, stride 2	$8\times8\times N$
	$8\times8\times N$	CNN ×2	$8\times8\times N$
	$8\times8\times N$	upsampling (bilinear interpolation)	$16\times16\times N$
	$16\times16\times N$	CNN	$16\times16\times N$
	$16\times16\times N$	GCN	$1\times1\times N$
Spectrum reconstruction	$13\times13\times k$	basic module	$13\times13\times k$
	$13\times13\times k$	basic module	$13\times13\times N$
	$13\times13\times N$	basic module ×2	$13\times13\times N$

TABLE II
CLASSIFICATION PERFORMANCE OF THE COMPARED BS METHODS ON THE SALINAS DATASET

Data	Methods	SVM	EPF	LCMV	2DCNN
Salinas	Full bands	91.26±0.12	95.77±0.55	96.37	92.21±3.86
	UBS	89.73±0.35	94.80±0.65	96.49	92.24±1.10
	MEAC	90.57±0.35	95.79±0.47	96.45	92.52±1.72
	MDPP	90.32±0.54	95.86±0.79	96.25	92.96±0.95
	LRR-BS	91.34±0.80	96.10±0.62	96.62	92.76±1.49
	ASPS	90.06±0.19	94.66±0.25	**97.09**	92.98±0.76
	TRC-OC-FDPC	90.17±0.23	96.13±1.07	96.35	93.01±0.91
	GCN-CNN-BS	**91.35±0.10**	**96.15±0.30**	96.97	**93.58±0.45**

2) Pavia University Dataset: The Pavia University dataset was captured by Pavia University, Italy, with the ROSIS-03 sensor in 2001. It consists of 610 × 340 pixels with a spatial resolution of 1.3 m × 1.3 m and has 103 spectral bands in the wavelength range from 0.43 to 0.86 μm after eliminating 12 bands due to the influence of noise, and it contains nine classes.

3) Houston Dataset: The Houston dataset was obtained from the ITRES CASI-1500 sensor and provided by the 2013 IEEE GRSS Data Fusion Competition. The data size is 349 × 1905, including 144 bands of spectral range from 364 to 1046 nm and 15 land-cover classes.

The colorful ground-truth images of the Salinas, Pavia University, and Houston datasets are displayed in Figs. 4–6.

B. Experimental Configuration and Compared Methods

All experimental results are implemented on a computer with a 64-bit Windows10 system, which is configured with Intel[1] Xeon,[1] CPU E5-2650 v2 @ 2.60 GHz, 56-GB operating memory, and MATLAB2016a simulation platform. Python language and PyTorch library are adopted in our approach. In this article, we employ the virtual dimensionality (VD) [42] algorithm to determine the intrinsic dimension, which is the number of selected bands for HSIC. The value of k is 21 for Salinas, 14 for Pavia University, and 22 for Houston data, respectively. This article compares our method with other state-of-the-art approaches, such as the uniform band selection (UBS), minimum estimated abundance covariance (MEAC) [43], MDPP [7], the original LRR model [44], ASPS [45], and TRC-OC-FDPC [46] on three publicly available HSI datasets to verify the feasibility of the proposed method. For the UBS method, the interval band of the Salinas is set to 11, 8 for the Pavia University, and 6 for Houston correspondingly. This article employs edge-preserving filtering (EPF) [47], linearly constrained minimum variance (LCMV) [48], SVM [49], and the 2DCNN method [6] for different classifiers to verify the classification of the selected band subset. To ensure a reliable classification result, we conduct each method five times with overall accuracy (OA) computed and randomly select the training set and the test set during each time.

[1]Registered trademark.

C. Design of Experiments

In the experiment, the index of the selected bands with the different methods is listed in Table V. Subsequently, to prove the effectiveness of the selected subsets, we fed all the band subsets selected by different BS methods that are displayed in Table V into four different classifiers separately. On the four types of classifiers, the numbers of classes are set as 16, 9, and 15 for the Salinas dataset, the Pavia University dataset, and the Houston data, respectively, and the numbers of bands are fixed to 21, 14, and 22 according to the VD algorithm. The LCMV employed the Gaussian filter with σ of 0.5 and the window size of 5, and the stopping threshold is 0.99. The SVM classifier

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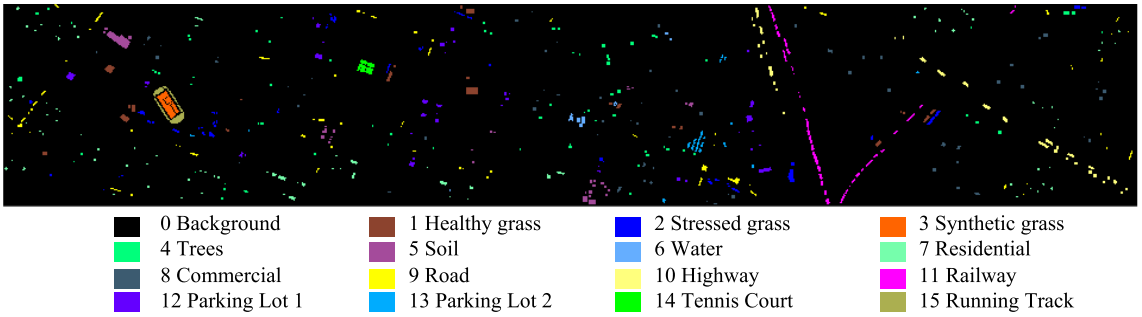


Fig. 6. Colorful ground-truth images of houston.

TABLE III
CLASSIFICATION PERFORMANCE OF THE COMPARED BS METHODS ON THE PAVIA UNIVERSITY DATASET

Data	Methods	SVM	EPF	LCMV	2DCNN
Pavia University	Full bands	**91.86±0.59**	98.49±0. 28	84.32	96.63±0.73
	UBS	86.45±0.73	97.62±0.79	84.98	97.71±0.97
	MEAC	78.01±1.16	91.34±0.87	83.98	94.89±1.86
	MDPP	83.93±0.48	95.28±1.32	84.43	95.82±0.98
	LRR-BS	82.10±0.80	93.60±2.20	84.15	96.78±0.60
	ASPS	82.57±0.49	95.40±0.31	83.36	97.66±0.15
	TRC-OC-FDPC	86.85±1.15	97.19±1.69	**85.79**	97.72±0.69
	GCN-CNN-BS	90.17±0.19	**98.72±0.13**	85.58	**97.86±0.53**

TABLE IV
CLASSIFICATION PERFORMANCE OF THE COMPARED BS METHODS ON THE HOUSTON DATASET

Data	Methods	SVM	EPF	LCMV	2DCNN
Houston	Full bands	67.04±0.40	82.55±1.00	63.58	90.67±0.82
	UBS	65.65±1.33	79.64±1.48	62.65	88.88±0.24
	MEAC	65.39±1.35	82.19±2.17	60.99	90.46±0.85
	MDPP	66.84±0.81	82.76±1.48	62.34	89.68±0.79
	LRR-BS	64.83±1.40	81.95±3.70	60.91	88.02±0.79
	ASPS	65.46±0.89	83.04±1.82	**66.16**	90.85±0.78
	TRC-OC-FDPC	**67.39±0.87**	82.12±1.31	64.15	90.39±0.45
	GCN-CNN-BS	67.34±0.43	**84.46±0.59**	63.99	**93.09±0.30**

adopts a Gaussian RBF kernel. The parameter settings of the 2DCNN architecture include the dropout rate of 0.6, the patch size of the input data of 11×11, the batch size of 100, and the iteration number of training of 4000. Our architecture fixes the alpha to 0.1, and the learning rate and the training epochs are set to 0.001 and 200, respectively. Besides, the Adam optimizer is adopted in the training process in pursuit of robustness. In the experiment for all the three datasets, the SVM, EPF, and 2DCNN methods are implemented with 5% of training samples chosen randomly from the training set. The input of the LCMV method is the spectral signature of the specific class computed with the label. Our proposed network is fed with a sample size of 13×13 on three datasets as input of the HSI cube. The specific structure of the FE and SR module is illustrated in Fig. 2, and the hyperparameter settings of the proposed GCN-CNN-BS approach are shown in Table I.

D. Results and Analysis

1) Performance Comparison of Different Methods: We conduct exhaustive experiments to validate the effectiveness of the proposed method, which compares the classification performance of different BS methods under different band subsets. Table II illustrates the classification performance of the compared BS methods on four classifiers with the Salinas dataset; it can be observed that the proposed GCN-CNN-BS approach performs best in SVM, EPF, and 2DCNN classifiers compared with other BS methods for HSIC, and our method even achieves better performance than the full bands. On the three classifiers of SVM, LCMV, and 2DCNN, the results of our BS method are similar to the results of the approach with full bands. In particular, for the EPF, the result of our method achieves the best OA of 96.15%, which increases by 0.38% over the full band, which generates an OA of 95.77% in all comparison methods.

Table III demonstrates the experimental results of different methods on the Pavia University dataset; it can be observed that the proposed GCN-CNN-BS approach performs best on the EPF, LCMV, and 2DCNN classifiers compared with other BS methods for HSIC; and the proposed method achieves OA of 98.72%, 85.58%, and 97.86%, respectively. For the SVM, the result of our method is slightly worse than the full bands and better than the rest of the methods. Specifically, among

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TABLE V
CLASSIFICATION PERFORMANCE OF DIFFERENT MODELS ON THREE DATASETS

Data	Methods			Classifier			
	CNN	GCN	GAT	SVM	EPF	LCMV	2DCNN
Salinas	✓	✓		**91.35±0.10**	**96.15±0.30**	**96.97**	**93.58±0.45**
	✓			88.38±0.19	94.99±0.54	96.86	91.75±0.24
	✓		✓	84.73±0.07	94.78±0.93	96.78	90.71±0.98
Pavia University	✓	✓		**90.17±0.19**	**98.72±0.13**	**85.58**	**97.86±0.53**
	✓			81.85±0.26	93.80±0.42	84.70	91.37±0.46
	✓		✓	87.53±0.71	97.65±0.83	85.03	95.14±0.28
Houston	✓	✓		**67.34±0.43**	**84.46±0.59**	**63.99**	**96.09±0.30**
	✓			65.34±0.34	81.82±0.93	62.96	95.15±0.23
	✓		✓	61.20±0.35	82.27±1.66	61.75	94.92±0.20

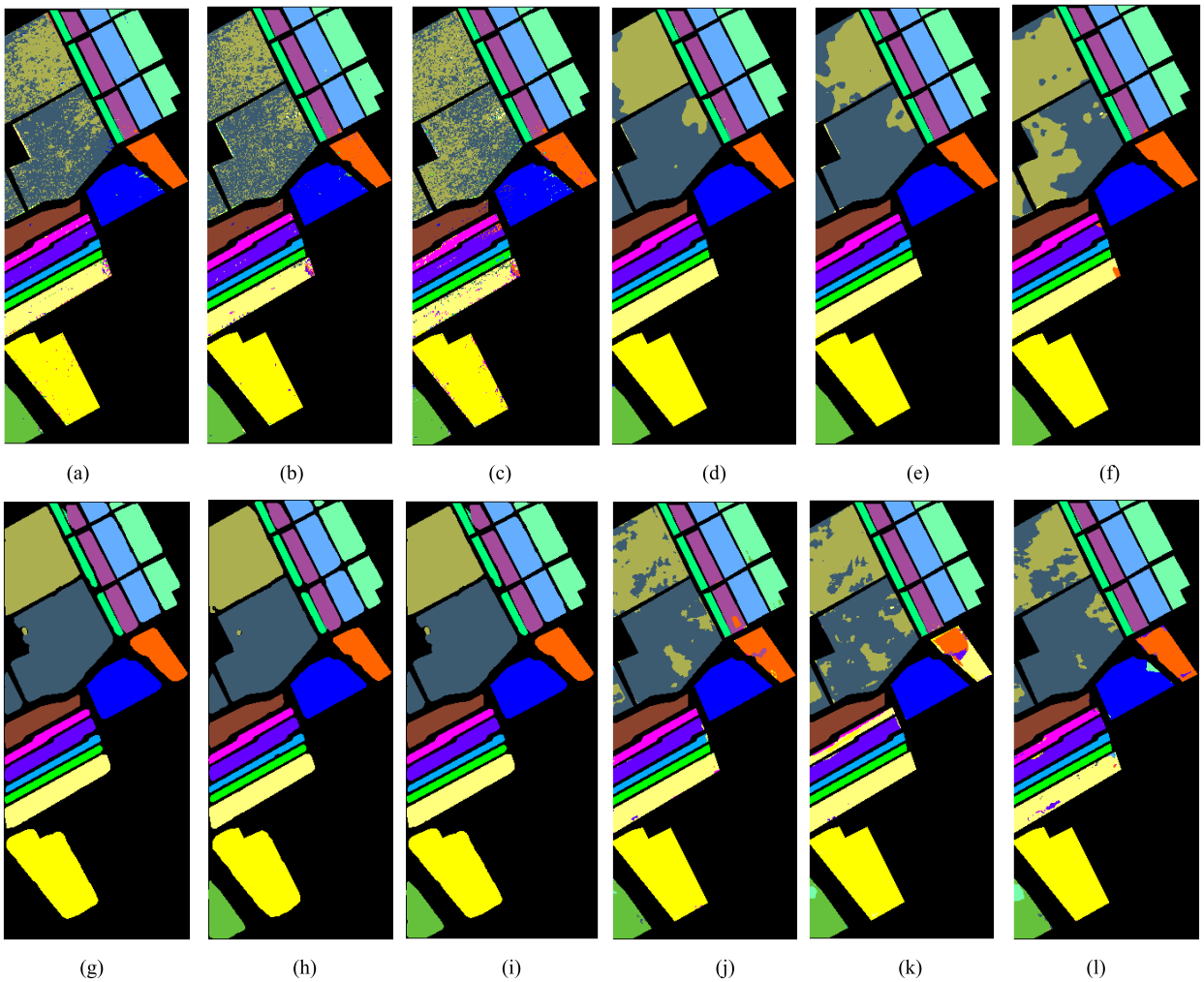


Fig. 7. Classification maps obtained by different variant versions of the proposed BS methods with different classifiers on the Salinas dataset. (a) SVM: GCN-CNN-BS (OA = 91.35%). (b) SVM: CNN-BS (OA = 88.38%). (c) SVM: GAT-CNN-BS (OA = 84.73%). (d) EPF: GCN-CNN-BS (OA = 96.15%). (e) EPF: CNN-BS (OA = 94.99%). (f) EPF: GAT-CNN-BS (OA = 94.78%). (g) LCMV: GCN-CNN-BS (OA = 96.97%). (h) LCMV: CNN-BS (OA = 96.86%). (i) LCMV: GAT-CNN-BS (OA = 96.78%). (j) 2DCNN: GCN-CNN-BS (OA = 93.58%). (k) 2DCNN: CNN-BS (OA = 91.75%). (l) 2DCNN: GAT-CNN-BS (OA = 90.71%).

the remaining comparison methods except for the full bands, the best classification result is TRC-OC-FDPC, which leads to the best OA of 86.85%, while our method result is 90.17%, which is significantly better than TRC-OC-FDPC.

Table IV presents the experimental results of different methods on the Houston dataset; it can be observed that the proposed GCN-CNN-BS approach achieves the best performance in the EPF and 2DCNN classifiers compared with

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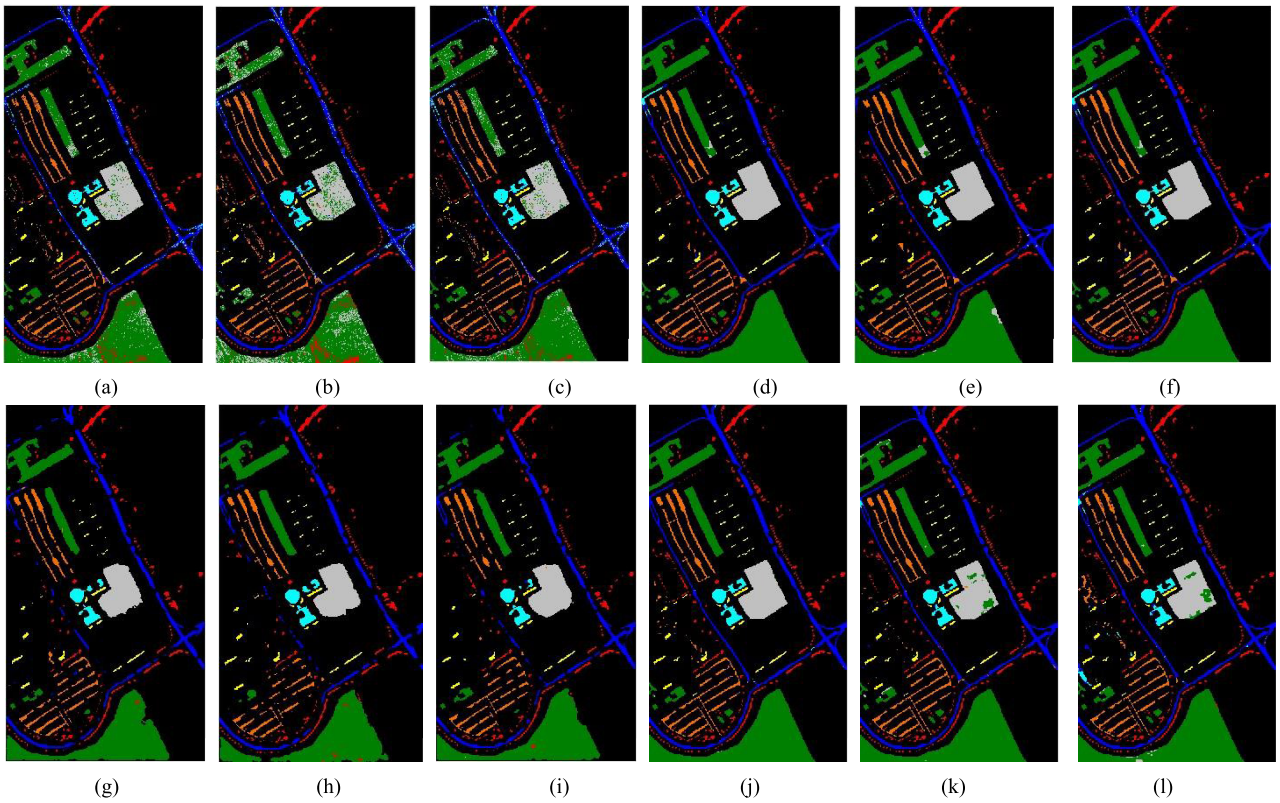


Fig. 8. Classification maps obtained by different variant versions of the proposed BS methods with different classifiers on the Salinas dataset. (a) SVM: GCN-CNN-BS (OA = 90.17%). (b) SVM: CNN-BS (OA = 81.85%). (c) SVM: GAT-CNN-BS (OA = 87.53%). (d) EPF: GCN-CNN-BS (OA = 98.72%). (e) EPF: CNN-BS (OA = 93.80%). (f) EPF: GAT-CNN-BS (OA = 97.65%). (g) LCMV: GCN-CNN-BS (OA = 85.58%). (h) LCMV: CNN-BS (OA = 84.70%). (i) LCMV: GAT-CNN-BS (OA = 85.03%). (j) 2DCNN: GCN-CNN-BS (OA = 97.86%). (k) 2DCNN: CNN-BS (OA = 91.37%). (l) 2DCNN: GAT-CNN-BS (OA = 95.14%).

different BS methods in terms of OA, which demonstrates the effectiveness of the proposed method. For SVM, TRC-OC-FDPC presents the best result OA of 67.39%, while the OA of GCN-CNN-BS is 67.34%. There is almost no difference between the results of TRC-OC-FDPC and GCN-CNN-BS. In particular, for the EPF and 2DCNN, the OA of our method is 84.46% and 93.09%, which increases by 1.91% and 2.42% over the full bands, which generates the best OA of 82.55% in all comparison methods.

2) Analysis of Module Function: In addition, to further explore the robustness of our proposed method, we also set up a series of ablation experiments. First, we eliminate all GCNs in the model to ensure that only CNN remains, namely, CNN-BS. Second, we replace all GCN models with GAT [50], namely, GAT-CNN-BS. Experiments verify that the results obtained after changing the model are inferior to the original model.

Therefore, the experiments prove that our proposed method is feasible. Figs. 7–9 show the classification maps obtained by different methods and different classifiers on the three datasets. Table V shows the performance of different models on the three datasets. First, we take the CNN-BS method compared with our method to demonstrate the usefulness of our method.

Although CNN can excavate the spatial correlation, it is unable to fully exploit the rich spectral information of the HSI, and the feature images input to the following network only contains spatial information but without spectral information; as a result, the subsequent spectral reconstruction module cannot receive spectral information and completely restore the original images. The data listed in Table V indicates that our architecture greatly improved the classification performance on the three datasets trained by GCN and CNN.

On the Salinas dataset, the OAs of CNN-BS on SVM, EPF, LCMV, and 2DCNN are 88.38%, 94.99%, 96.86%, and 91.75%, respectively, which are 2.97%, 1.16%, 0.11%, and 1.83% lower than GCN-CNN-BS. On the Houston dataset, the CNN-BS achieves classification results with the OA of 65.34%, 81.82%, 62.96%, and 95.15% on the SVM, EPF, LCMV, and 2DCNN classifiers, while the impact on the LCMV classifier OAs is least on those two datasets.

The classification performances of CNN-BS is evaluated on the Pavia University dataset are 81.56%, 93.80%, 84.70%, and 91.37% on SVM, EPF, LCMV, and 2DCNN are, respectively, which are 8.32%, 4.92%, 0.88%, and 6.49% lower than GCN-CNN-BS. The specific data listed in Table V indicate that the removal of the GCN module has the greatest impact on the SVM classification; on the contrary, it has the least impact on the LCMV classification.

GCN and GAT are multilayer neural networks that operate on a graph and generate node embeddings by gradually fusing the features in the neighborhood. The difference between them is that GCN performs eigen-decomposition on the Laplacian matrix, and GAT implements a self-attention attention

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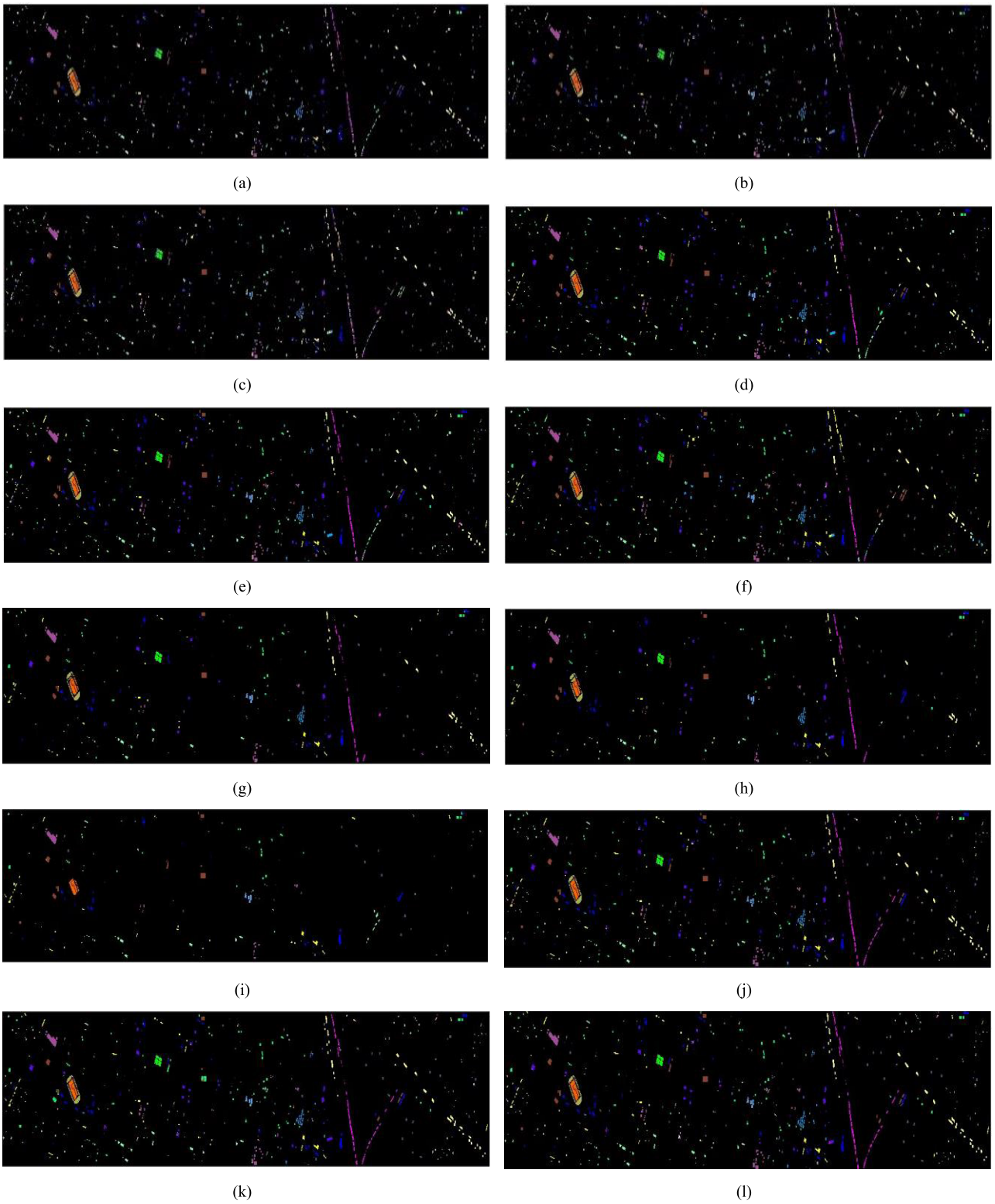


Fig. 9. Classification maps obtained by different variant versions of the proposed BS methods with different classifiers on the Salinas dataset. (a) SVM: GCN-CNN-BS (OA = 67.34%). (b) SVM: CNN-BS (OA = 65.34%). (c) SVM: GAT-CNN-BS (OA = 61.20%). (d) EPF: GCN-CNN-BS (OA = 84.46%). (e) EPF: CNN-BS (OA = 81.82%). (f) EPF: GAT-CNN-BS (OA = 82.27%). (g) LCMV: GCN-CNN-BS (OA = 63.99%). (h) LCMV: CNN-BS (OA = 62.96%). (i) LCMV: GAT-CNN-BS (OA = 61.75%). (j) 2DCNN: GCN -CNN-BS (OA = 96.09%). (k) 2DCNN: CNN-BS (OA = 95.15%). (l) 2DCNN: GAT-CNN-BS (OA = 94.92%).

mechanism for each node. Next, we compare the classification results of GAT-CNN-BS with GCN-CNN-BS.

Table V demonstrates the results of GAT-CNN-BS, in which we compare our method with GAT-CNN-BS in terms of the significance of the difference between their classification results. In general, the improvement of accuracies yielded by our approach instead of GAT-CNN-BS for all the datasets and GAT combines graph convolution with self-attention mech-

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TABLE VI
INDEX OF THE SELECTED BANDS BY THE COMPARED BS METHODS

Data set	BS methods	Index of the selected bands
Salinas (21)	UBS	1, 12, 23, 34, 45, 56, 67, 78, 89, 100, 111, 122, 133, 144, 155, 166, 177, 188, 199, 210, 224
	MEAC	3, 5, 8, 10, 12, 17, 18, 25, 28, 32, 36, 44, 51, 58, 68, 105, 107, 110, 148, 149, 203
	MDPP	1, 8, 11, 22, 27, 28, 50, 57, 58, 65, 90, 99, 105, 119, 123, 134, 142, 157, 175, 191, 204
	LRR-BS	4, 11, 13, 21, 22, 35, 41, 47, 51, 53, 65, 66, 70, 86, 98, 102, 103, 104, 132, 173, 175
	ASPS	15, 30, 31, 46, 53, 54, 67, 90, 93, 105, 112, 123, 129, 151, 153, 163, 185, 190, 196, 202, 224
	TRC-OC-FDPC	7, 11, 14, 20, 24, 32, 37, 39, 40, 51, 55, 57, 59, 62, 64, 68, 75, 82, 88, 96, 109
	GCN-CNN-BS	39, 57, 59, 60, 69, 79, 80, 84, 90, 95, 103, 119, 127, 128, 144, 151, 161, 186, 187, 198, 208
Pavia University (14)	UBS	1, 9, 17, 25, 33, 41, 49, 57, 65, 73, 81, 89, 97, 103
	MEAC	1, 23, 24, 25, 31, 40, 42, 47, 48, 54, 56, 58, 59, 83
	MDPP	2, 23, 44, 46, 50, 62, 66, 73, 89, 91, 92, 93, 96,102
	LRR-BS	6, 10, 13,16, 18, 23, 28, 33, 74, 79, 88, 90, 93, 98
	ASPS	13, 16, 23, 34, 36, 37, 39, 40, 41, 47, 84, 89, 94, 103
	TRC-OC-FDPC	4, 15, 19, 29, 33, 41, 48, 53, 61, 67, 72, 80, 88, 99
	GCN-CNN-BS	5, 14, 26, 56, 72, 78, 80, 83, 84, 86, 88, 90, 94, 98, 99
Houston (22)	UBS	1, 7, 13, 19, 25, 31, 37, 43, 49, 55, 61, 67, 73, 79, 85, 91, 97, 103, 109, 115,121, 127
	MEAC	1,2,4,7,17,18, 20, 26, 32, 42, 50, 51, 57, 61, 66, 68, 71, 77, 82, 89, 110, 130
	MDPP	8, 12, 19,43, 50, 51, 57, 63, 67, 70, 78, 86, 89, 102, 126, 108, 113, 115, 120, 123, 131, 133
	LRR-BS	7, 10, 11,31, 32, 33, 34, 52, 57, 59, 60, 77, 89, 86, 102, 117, 118, 122, 123, 126, 130, 143
	ASPS	2, 13, 18, 26, 27, 42, 44, 57, 60, 68, 69, 79, 88, 91 ,101, 105, 110, 116, 126,127, 135, 137
	TRC-OC-FDPC	2, 5, 15, 20, 24, 36, 40, 48, 59, 63, 71, 74, 78, 90, 98, 104, 120, 121, 131, 134, 141, 143
	GCN-CNN-BS	2, 4, 9, 16, 37, 39, 40, 49, 67, 79, 88, 98, 106, 108, 120, 121, 123, 126, 127, 136, 138, 142

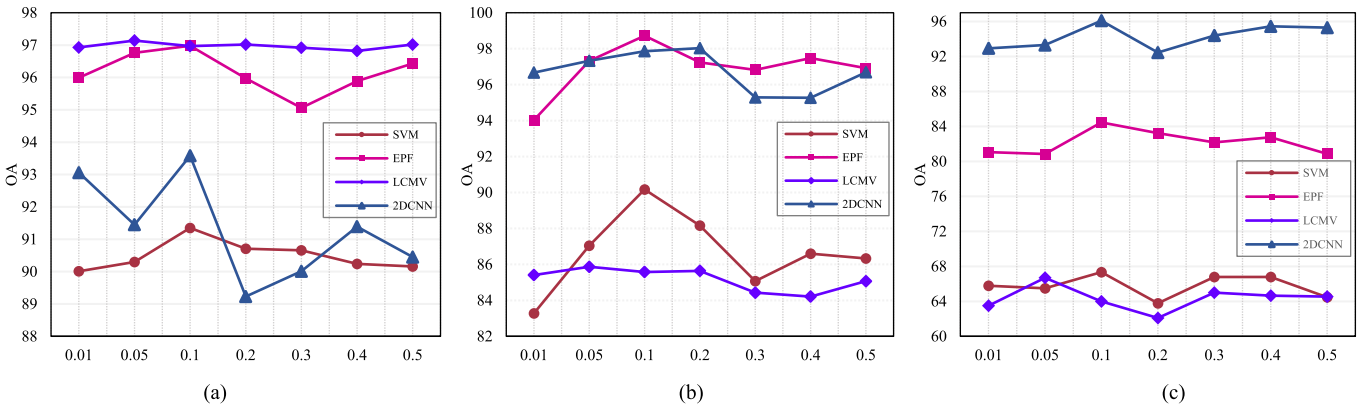


Fig. 10. Influence of α with different values on the classification results. (a) Salinas dataset. (b) Pavia University dataset. (c) Houston dataset.

anisms by emphasizing the importance of different neighbor nodes. However, since HSI is composed of hundreds of continuous narrow bands with high correlation, the high correlation between bands may lead to poor performance of the self-attention mechanism of GAT. Figs. 7–9 exhibit a visual comparison of the classification results generated by GCN-CNN-BS, CNN-BS, and GAT-CNN-BS on the datasets that we employed. It can be seen that the misclassified pixels exist differences in each dataset; in comparison with other methods, our architecture provides fewer misclassified pixels in different classifiers. In summary, our method provides competitive classification performance from different perspectives.

3) Analysis of Parameter Settings of Adjacent nodes: The proposed method is achieved with the structural graph obtained from a mass of hyperspectral bands. To investigate how the performance can be influenced by the different ways of constructing a graph, we set up the comparison experiment of different α's, where $r_{(v_i,v_j)}$ obeys the Bernoulli distribution with the parameter of α. In our experiments, α is set to 0.01, 0.05, 0.1, 0.2, 0.3, 0.4, and 0.5, and these parameters represent the percentage of connections between nodes. The specific experimental results are shown in Fig. 9.

In Fig. 10, we find that all the three datasets have achieved the best classification performance on SVM and EPF; when $\alpha = 0.1$, the best OA is yielded with $\alpha = 0.05$ on the LCMV classifier. For the Salinas dataset, SVM and EPF produce the worst OAs, which are 90.01% and 95.06%, when $\alpha = 0.01$ and α =0.3, respectively. 2DCNN provides the best result with the setting of $\alpha = 0.1$ and the worst result at α =0.2; as can be seen from Fig. 10(a), the different parameters that we set have a significant impact on the classification results of 2DCNN. Relatively speaking, different parameters have minimal impact on the classification results of the LCMV classifier.

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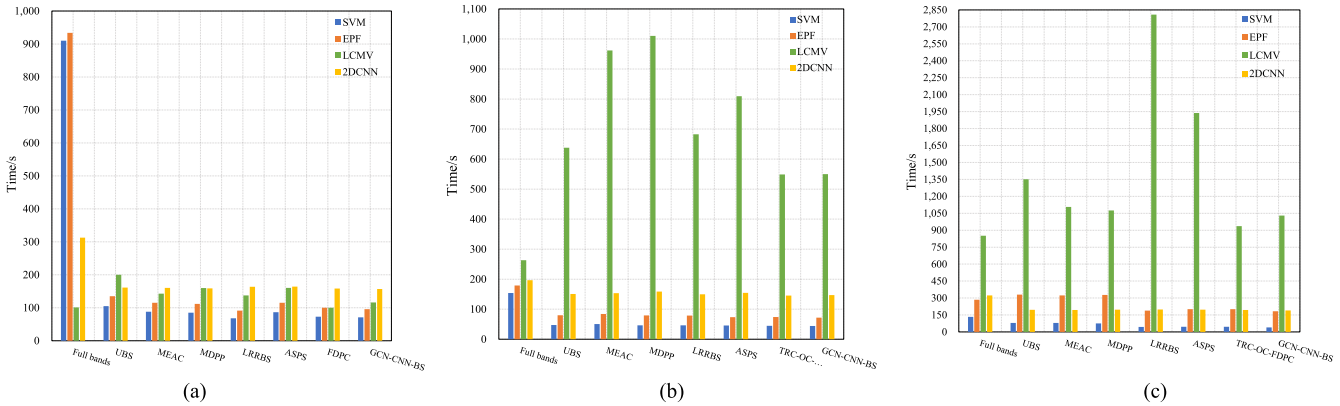


Fig. 11. Classification time cost with different BS methods. (a) Salinas dataset. (b) Pavia University dataset. (c) Houston dataset.

For the Pavia University dataset, the best OA of the SVM classifier is 90.17% when $\alpha = 0.1$, while the worst OA of the SVM classifier is 83.28% when $\alpha = 0.01$, 6.89% lower than the best result. Similarly, on the EPF classifier, when $\alpha = 0.01$, the OA of the SVM classifier is 94.03%, which is the worst result; it indicates that the connections between nodes are too sparse to permit GCN to aggregate the information of neighbor nodes. From Fig. 10(b), we can see that different parameters have the greatest influence on the classification results of the SVM classifier; they have little influence on OA in the LCMV classifier. 2DCNN achieves satisfactory performance at $\alpha = 0.2$. The change of parameters makes the OAs of SVM, LCMV, and 2DCNN classifiers on the Houston dataset fluctuate greatly, while the EPF is relatively stabilized.

4) Classification Time Performance: The running time of the GCN-CNN-BS with a comparison of the different classifiers is shown in Fig. 11(a)–(c). It can be clearly seen from the figure that, on the three classifiers of SVM, EPF, and 2DCNN, for the three datasets, our proposed BS method takes much significantly less time than the full bands, which is fully indicates that BS methods can effectively reduce the information redundancy of HSI and the running time cost; as a result, it is of great significance to the processing of HSI.

The LCMV classifier spends more time with the selected bands owing to the characteristics of the LCMV classifier. For the Salinas dataset, our method runs almost as fast as the full bands. On the Pavia University dataset and the Houston dataset, our method consumes more time than the full bands, but other comparison methods also run slower than full bands. It indicates that our methods are more competitive than the compared BS methods.

IV. Conclusion

In this article, we developed a novel hybrid GCN for self-supervised BS for HSIC. This approach adopts spectrum rebuilding as a self-supervised learning task and substitute for the complex searching of subset paths. To encode the intrinsic relationship among bands reliably, we employ graph convolution for both the topological organization and the interactive information extraction of the bands. Specifically, we regard hyperspectral bands as nodes to construct the graph structure, and the band correlation is mined deeply by a hybrid network composed of GNN and GCN modules, which fully exploits the spectral–spatial information locally and globally. Therefore, the relations of bands are gradually refined along with GCN and CNN. The experimental results on three widely used HSI datasets demonstrate that the proposed BS method for HSIC is capable of generating completive performance compared with the state-of-the-art methods. Since this article employs a relatively simple loss function to implement the self-supervised scheme, in future work, we will consider designing more diversified loss functions integrated with specific applications to explore the potential of our proposed framework.

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